

PERMARUN- A Persuasive Game to Improve User Awareness and Self-Efficacy Towards Secure Smartphone Behaviour

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ABSTRACT

Android smartphones have undergone various changes and, uniformity in design (both hardware and software) has become common in most phones. Yet, security and privacy issues persist and grow along with the evolution of smartphones. Recent research shows that user awareness is still low, and they do not follow secure behaviour while using smartphones. Although a handful of works exist for improving user awareness and self-efficacy, none of them educate the users about Android Permissions in a contextual manner. In this paper, we discuss our persuasive game - "*PermaRun*", which teaches and motivates users to follow secure smartphone behaviour, increases user awareness and self-efficacy about android permissions. We conducted a Heuristic Evaluation for Playability (HEP) and Persuasiveness Evaluation (PE). The result shows that players had a positive experience playing the game, and they found the game playable and persuasive.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Heuristic evaluations.

KEYWORDS

Secure Smartphone Behaviour, Persuasive Game, User Awareness, Self-Efficacy, Heuristic Evaluation for Playability

ACM Reference Format:

Anirudh Ganesh, Chinenye Ndulue, and Rita Orji. 2021. PERMARUN- A Persuasive Game to Improve User Awareness and Self-Efficacy Towards Secure Smartphone Behaviour. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451781>

1 INTRODUCTION

In these recent times, the technology underneath smartphones has undergone rapid changes, including hardware, operating system, software, and security improvements. Unfortunately, malware seems to have evolved along with these improvements. Humans are the weakest link in the chain of cybersecurity [6] and, they

have been ignored in the initial stages of the evolution of cybersecurity. Recent researches suggest that users are not aware of secure smartphone behaviour and other privacy implications [4, 6, 14, 28]. Google strives to protect users' data in various ways and, the most common way that has been around for a long time is the Android Permissions. Runtime permissions are used by Google [10], where the apps request permissions before accessing users' data. Explaining the reason for requesting permissions (called permission rationale) is optional for the app developers [11], and in the absence of an explanation, the user can only decide using the context of the action that triggered the permission request. Users often make wrong decisions out of impulse [5], ignoring what granting of permission would entail and, do not always check the permissions and the app's functionality [3]. In this paper, we discuss our persuasive game - "*PermaRun*" that we developed to educate the users about secure smartphone behaviour and improve awareness and self-efficacy about android permissions. We achieve this by providing tips about how to make secure smartphone decisions and secure smartphone behaviour throughout the game using interesting game narratives and persuasive strategies. We maintain the context of Android permissions with the help of a contextual story. We use nine persuasive strategies [21] from the Persuasive System Design Framework [20] to motivate the users and implement six game design patterns [13] to keep the game interesting.

2 BACKGROUND

Persuasive games has been designed to solve problems across different domains [12, 17, 25, 26]. However, cybersecurity games that educate people about smartphone security practices are few. Most of the existing games focus on other cybersecurity issues, such as detecting phishing links, spam emails and setting strong passwords [7, 9, 18, 29, 32, 33]. Ndulue et al. designed a match-making game called "*Phisher-Crush*" [18] to educate players about how to identify phishing links. "What.Hack" [33], a game that implemented situated learning, was designed to improve the self-efficacy of users, where the player takes the role of a phishing expert to identify phishing emails for a bank. Chen et al. [7] designed "*Hacked Time*", a role-playing game that improves self-efficacy, where the player takes the role of a detective to solve a case of password theft. For smartphone security, we were able to find only a handful of games. This might be because the topic is relatively new compared to other areas of cybersecurity.

"*HappyPermi*" [1] is an Android app that displays the permissions requested by other apps installed in the users' phone, along with users' data that are retrieved according to the permissions granted by the user for apps installed in their phone. During the post-study evaluation, users were surprised to see their data being displayed

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CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451781>

by the app and, most of them were concerned about their photos and contacts being accessible to the apps and blocked contacts permission after using “HappyPermi”. “Make my phone secure” [2], a role-playing game, where the player plays the role of an IT security expert and helps other non-playable game characters (NPC) with their smartphone security problems that revolve around Android permissions. The player must toggle appropriate permissions for apps in the game to provide or deny access to user data according to the problem statement presented to them in-game. Zargham et al. [35] designed a humorous game to educate users about smartphone security and privacy. The author conducted a study with a between-subject design, using the game, a humorous video and a serious video explaining smartphone security and privacy issues. The participants preferred the humorous game followed by the humorous video. However, the users also pointed out that once the player uncovers all the decisions, the game becomes boring.

We collectively refer to the general etiquettes and standards that are to be followed while using a smartphone to stay safe and protect oneself from malware and other kinds of security threats as Secure smartphone behaviour [3, 4, 6, 14, 28, 30, 36]. While there is plenty of research around games for cybersecurity, seldom work has been done on creating contextual persuasive games to raise awareness around secure smartphone behaviour. Therefore, we present “PermaRun”, a mobile persuasive game that promotes secure smartphone behaviour and improves user awareness about android permissions. We implemented six game design patterns as collated by Khalifa et al. [13] to keep the game interesting for the player and, to maintain the context, we included a story that revolves around Android Permissions. We implemented nine persuasive strategies from the Persuasive System Design model (PSD) [20] to motivate the players.

3 PERMARUN DESCRIPTION

In this section, we present the following game details: game story, game design, gameplay, game design patterns and persuasive strategies implementations.

3.1 Game Story

For maintaining the context of Android Permissions, we weaved a story that involved Android permissions. We named our main game character Dillon, who went on a camping trip to a forest with his friends. Their plans were ruined by a troll who was angry with humans - the inventor of smartphones. The troll was a new Android user who was not aware of the implications of Android Permissions and was a victim of data theft. The troll kidnapped Dillon’s friends and trapped them at various parts of the jungle and promised to release them if Dillon helped the troll to choose appropriate Android Permissions.

3.2 Game Design

The game development followed an iterative user-centred design approach. We developed an initial prototype of the game using Unity Game Engine [38], free game assets [39–43], Proto.io [27] and received informal feedback from the target audience. Using the feedback, we refined the design and developed *PermaRun* using Unity Game Engine. In the game, the player must collect appropriate



Figure 1: Collectible Permission Symbols

permission symbols for each specific app. Throughout the game, we operationalized Android runtime Permission Symbols as in-game collectibles and, towards the end of the game, if the player manages to collect all the necessary permissions, they would be able to save Dillon’s friends. We discuss the details of the game in the upcoming section.

3.3 Gameplay

In the current implementation of the game, the player must collect appropriate permission symbols for different types of apps - for instance, an online music player app. We include permissions that are essential for the functionality of an app along with permissions that are not necessary for the app. For example, an online music player app might require location permission to retrieve the users’ current location and recommend local content to the user, or the app might require microphone permission for a voice search feature. The necessary permission symbols are coded with green colour, whereas the unnecessary permission symbols are red in colour. We highlight the app features before the beginning of a level, and we present the players with a list of permission symbols as a pre-game quiz from which they must select the appropriate permissions, which they think, are required for the app considering its functionality.

We have colour-coded permissions as in-game collectibles in a traffic signal format (Figure 1). We also have enemies to make the game interesting and challenging. When the player destroys the enemies or collects a necessary permission symbol, their score increases. When they pick up an unnecessary permission symbol, their score decreases, and their movement speed reduces for a short period. To measure the players’ knowledge and the games ability to facilitate learning, we included a pre-game question, and we ask the same question as a post-game quiz. We provide feedback for the player about their responses at the end of the game session – displaying their scores along with the feedback explaining which permissions are necessary and which are not. The details of the implementation of the persuasive strategies and the game design pattern are shown in (Table 1) and (Table 2).

3.4 Implementation of Game Design Patterns and Persuasive Strategies

We implemented six game design patterns as collated by Khalifa et al. [13] to make the game interesting and nine persuasive strategies as proposed by Oinas-kukkonen et al. [20] to motivate the players.



Figure 2: Permission Symbols description before the start of the game (Left), In-Game Suggestions (Right)

Table 1: Implementation of Game Design Patterns

Game Design Pattern	Description	Implementation
<i>Branching</i>	Giving the player multiple paths to explore in-game is classified as branching. If we apply constraints, where the player needs to meet specific conditions to choose another pathway, it is called conditional branching [13].	We implement both branching and conditional branching. For branching, we give an alternate path to the player for exploration, and for conditional branching, the player is teleported to another location if they pick up a gem. This makes the game more interesting and increases the curiosity of the player.
<i>Guidance</i>	A non-verbal game element that guides the player towards the goal or in the right direction is classified as guidance [13].	We used pathways to guide the player. We placed enemies and collectibles to guide them towards a specific path.
<i>Foreshadowing</i>	Teasing the player with an in-game element with which they might not be able to interact with or can be a game element that changes over the course of the game comes under foreshadowing [13].	We tease the players with hidden areas or inaccessible game items that would be accessible as they progress through the game. We introduce enemies in a stationary way, followed by patrolling enemies at a later stage. This makes the game more interesting to play and might increase the players' curiosity [31].
<i>Safe Zone</i>	The areas in-game, where the player is not in danger and can think about their next actions, or where they can take a break from all the tension that was built in-game can be classified as safe zone [13].	For safe zones, we have checkpoints (Player restarts the game from a particular point if they lose) in our game, where the player is not in danger and would feel safe. We have three checkpoints in our game. The player can plan their next moves or can just rest before progressing in the game. This breaks the tension in the game.
<i>Layering</i>	Combining multiple game elements or reusing game elements to introduce challenges is classified as layering [13].	We placed enemies in-front of collectibles, positioned app permission symbols near enemies, placed obstacles in the pathway, and players can reach some areas only through a moving platform. This makes the game challenging.
<i>Pace Breaking</i>	Increasing or decreasing the tension in-game is commonly referred to as pace breaking. Pace breaking is usually achieved by introducing bosses or giving the player a break after long combats [13].	We implement pace breaking by introducing enemies and then by giving the player a brief break from combat. We mix the Layering game pattern with Pace Breaking by adding obstacles to the paths to keep the game interesting, and the players engaged.

4 HEURISTICS AND PERSUASIVENESS EVALUATION

We evaluated our game using the Heuristic Evaluation for Playability [8] to test the playability and usability of the game. We also assessed the persuasiveness of the persuasive strategies implemented in the game with respect to their ability to promote the desired secure smartphone behaviour. We evaluated the game with six evaluators, which is in line with the recommended number of evaluators

for heuristic evaluation [19] and considered the aggregate for better results [19]. The criterion for selecting evaluators was that they should be researchers in the field of Persuasive Technology and must be familiar with the PSD Model [20]. We used a questionnaire consisting of a validated scale to assess heuristics [8] and persuasiveness of the persuasive strategies along with their description, each of which had a 5-point Likert scale. The evaluators played the game for an average of about 20 minutes before filling up the

Table 2: Implementation of Persuasive Strategies

Persuasive Strategy	Description	Implementation
<i>Simulation</i>	Systems that help the users to observe the link between cause and effect immediately can be persuasive [20]	When the user picks up a necessary permission, their score increases, but when they pick up an unnecessary permission, their score decreases, and the player's movement speed reduces for a short duration.
<i>Competition</i>	A system that persuades users to adopt a target behaviour by leveraging their drive to compete can be persuasive [20, 24]	We rank the players according to their in-game score on a leaderboard. This score is affected by various factors like the type of permissions the player collects in-game, items that they pick up (E.g., Fruits), and the number of enemies they destroy.
<i>Rewards</i>	A system that rewards the user for performing a behaviour might be persuasive [20]	When the players pick up the correct permission or if they destroy an enemy, their score increases. When the player picks up fruits, their health increases. There are also extra lives in hidden areas for the player to collect.
<i>Suggestion</i>	A system that offers appropriate suggestions tend to be more persuasive [20]	We give immediate feedback and tips regarding secure smartphone behaviour (Figure 2). Whenever the player picks up a permission, we show suggestions about secure smartphone behaviour.
<i>Liking</i>	A system that is visually attractive to the user is more likely to be persuasive [20]	We added sound effects for the game character, background music followed by other in-game features and animations.
<i>Tunnelling</i>	A system that guides a user through a process or an experience and persuades them along the way [20]	We have three instances of tunnelling in our game – (a) before the start of the game, where the player gets an instruction screen that explains how to play the game. (b) In-game Collectibles to guide the player to the end of the level. (c) Providing information about the run-time permissions to the players to give them context about the permissions (Figure 2).
<i>Self-Monitoring</i>	A system that helps the user to keep track of their performance [20, 22]	The player has a Heads-Up Display (HUD) in between the in-game controls, which is displayed throughout the game that shows the necessary permissions that are needed to complete a level.
<i>Reduction</i>	A system that reduces the work load into simple tasks to help the users perform a target behaviour and might increase the benefit/cost ratio of a behaviour [20]	We consider in-game checkpoints as a reduction strategy because the player need not repeat the whole level if they make a mistake and lose a life in the middle of the game.
<i>Praise</i>	A system that praises the user for performing a target behaviour might be persuasive [20]	The users receive praise (in the form of text) if they get the right answers in the post-game quiz after playing the game.

HEP and Persuasiveness questionnaire. The Heuristic Evaluation of Playability [8] consists of forty-three (43) heuristics questions grouped into four categories based on the aspect of the game they assess: *Game Play*, *Game Story*, *Game Mechanics* and *Game Usability*. We will be discussing the results of HEP in terms of these four categories in the upcoming section.

4.1 Analysis of the Heuristic Evaluation for Playability Results

From the analysis of the 5-point Likert Scales and the feedback that we received for the Heuristic Evaluation for Playability [8], it was evident that *Game Mechanics* scored the highest (mean = 4.09, sd = 0.42) followed by *Game Usability* (mean = 4.02, sd = 0.41), *Game Play* (mean = 3.93, sd = 0.50) and *Game Story* (mean = 3.75, sd = 0.61). *Game Story*'s score is not bad because explaining the

importance of security to an end-user has always been a challenge, and it was above the neutral score of three (3) on a 5-point Likert Scale. All the game heuristics scored above the neutral score of three (3) (Figure 3) on the 5-point Likert Scale. Thus, from the results of the Heuristic evaluation for Playability [8], we conclude that *PermaRun* is playable.

4.2 Analysis of Persuasiveness Results

With respect to the persuasiveness of the strategies, all the strategies were found to be persuasive (higher than the neutral rating of 3) except *Praise* and *Competition*. Specifically, *Liking* (mean = 4.6, sd = 0.47) and *Suggestion* (mean = 4.5, sd = 0.5) scored the highest among all the strategies. *Praise* (mean = 2.8, sd = 1.86) scored the lowest among all the strategies (Figure 3). The score of the *Liking* strategy can be attributed to the game design patterns that we

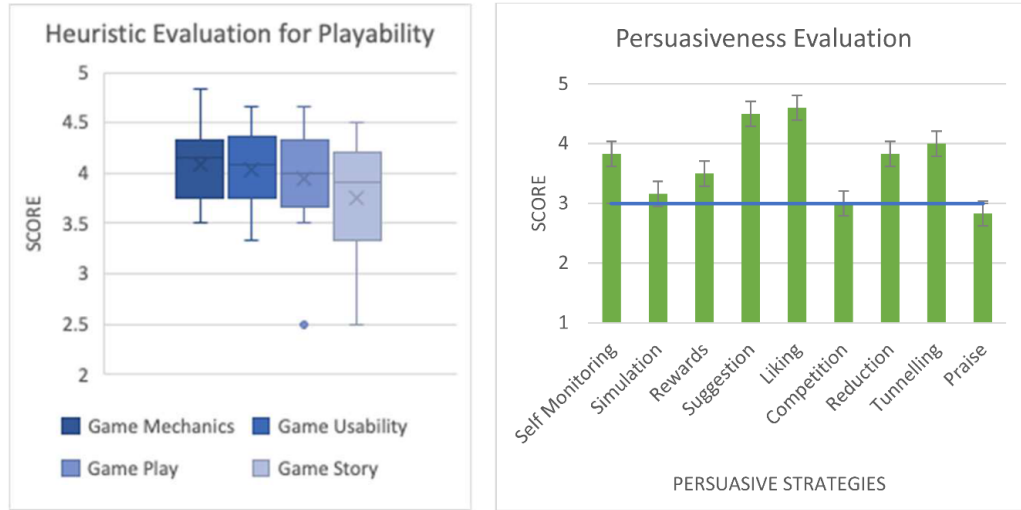


Figure 3: Box and Whisker Plot for Heuristic Evaluation for Playability Results (Left), Bar Chart for Persuasiveness Results (Right)

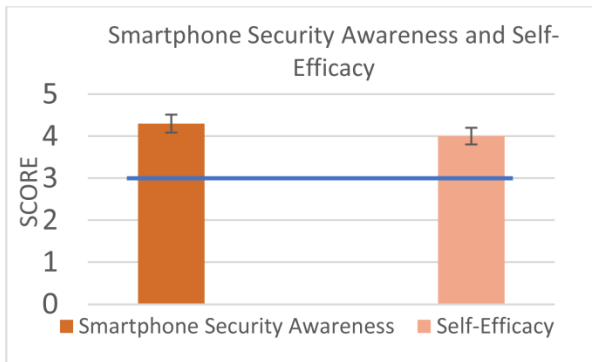


Figure 4: Bar Chart for Smartphone Security Awareness and Self-Efficacy

implemented and the attractive game assets that we used. The score of the *Suggestion* strategy can be attributed to the suggestions that we displayed when the player picks up a permission symbol which helps them to learn about safe smartphone security behaviour. We had only one instance of *Praise* towards the end of the game and, this might be a reason for the weak score.

4.3 Analysis of Evaluators Feedback

In this section, we discuss the analysis of evaluators' feedback. First, we discuss the results of smartphone security awareness and self-efficacy. Then, we discuss the analysis of their feedback about the game.

4.3.1 Smartphone Security Awareness and Self-efficacy. We asked the evaluators about their awareness and self-efficacy with respect to being able to adopt secure smartphone behaviour using a 5-point Likert Scale. Overall, they found the game *fun to play* and agreed that the game would improve the players' awareness (Mean = 4.3,

sd = 0.47) and self-efficacy (Mean = 4.0, sd = 0.81) to adopt secure smartphone behaviour. The mean scores were above the neutral point of three (3) on the 5-point Likert Scale (Figure 4).

4.3.2 Analysis of Qualitative Feedback. We collected qualitative feedback from the evaluators and performed thematic analysis using an affinity diagram and this helped us to uncover the common issues as well as what all the evaluators liked about our game. Some of the evaluators reported that they felt *nostalgic* when they started playing the game and a *sense of achievement* when they completed the game. One of the evaluators commented: "I could quickly relate it to mario" - E6. Recent research suggests that nostalgia is directly impacted by past memories and satisfaction of competence, especially with retro games that focus on challenging and fast gameplay [34]. Being able to relate the game with past experience is important for persuasive games since it reduces the learning curve and increases players feeling of competence in line with the Self-determination theory [34] and relevance with respect to being able to associate the game with past experience which is important to ensure players would play the game. This also confirms that our design decision to make the game play to resemble a well-known game *Super Mario* [44] to attract diverse audience and reduce learning curve paid off.

Most of the evaluators discovered at least one of the hidden areas in the game and, one evaluator found all three hidden areas in the game. The hidden areas create a sense of uncertainty or a knowledge gap that makes the user curious about visual cues and is called *perceptual curiosity* that leads to increased attention to the game [31]. Some evaluators felt that the *game was challenging to play*, and below are the reasons. One of the evaluators commented - "Considering the level has a timer, reading the tips is against the gameplay and I barely read even one tip to save time." - E3. The same evaluator also stated that "The HUD overlaps the map in many places, I died twice cause I couldn't see what I'm jumping at due to the interface on the bottom-middle" - E3. To solve this issue, we plan to

hide the HUD when the player performs a jump. Another evaluator stated - “While the player is collecting the icons (whether green or red), you can pause the gameplay, with an OK button, so that after the player reads the message, they can then press the OK button to resume the game” - E4. From this, we can correlate that **pausing the game while displaying suggestions** might be a useful feature. Ideally, games should pose some level of challenge to sustain players’ interest; if it is too easy, players may quickly get bored with the game in the long run. However, if persuasive games are too challenging, players may drop off before they are exposed to all the persuasive content that would lead to behaviour change. For persuasive games to be effective at impacting the desired changes, players need to be able to play and finish them. Therefore, we plan to use the feedback to improve our game and use adaptive game difficulty [37] to personalize the game challenge-level to match each player’s ability. This will in part ensure we are able to attract and sustain players’ interest.

5 CONCLUSION AND FUTURE WORK

We designed a persuasive game for improving users’ awareness and self-efficacy about secure smartphone behaviour. The key take-aways from our design are: (a) Our game gives instant feedback or tips about secure smartphone behaviour when the player picks up a permission symbol in-game. (b) We built the game according to six game design patterns [13] to make it more enjoyable for the player. (c) We impart skills for secure smartphone behaviour and expose the dangers of unnecessary permissions to an individual’s privacy and security using an interesting game narrative while maintaining the context of smartphone security. (d) We evaluated our game according to the Heuristic Evaluation for Playability [8] on six persuasive technology researchers and found that the game was playable with all the game heuristics scoring above the neutral score of three (3) on a 5-point Likert scale. The evaluators also found our game to be persuasive and feasible to increase users’ security awareness and self-efficacy about secure smartphone behaviour. For our future work, we plan to (i) Implement adaptive difficulty [37] in the game so that all players can complete the level and be engaged. (ii) Display the number of hidden areas discovered by the players towards the end of the level as this might increase their curiosity [31] to play again and explore more. (iii) Since the praise strategy got a low score due to few implementation instances, we plan to add praise in-between the game instead of just having it towards the end. (iv) We plan to include an audio narration for the story so that the players can follow along with the text easily. We are also planning to develop more game levels for a variety of apps permissions and we are considering tailoring the game for various playstyles [15, 16, 23]. Finally, we hope to conduct a long-term user evaluation to assess the effectiveness of the game on Smartphone users.

ACKNOWLEDGMENTS

This research was undertaken, in part, thanks to funding from the Canada Research Chairs Program. We acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC) through the Discovery Grant.

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